



BACHELOR OF SCIENCE IN INFORMATICS & COMPUTER SCIENCE
CAT 1

ICS 4111: Embedded Systems and Internet of Things

DATE: 21st May 2025

Venue: STMB-05-01

TIME: 1 Hour

INSTRUCTIONS

1. Answer all questions in this paper in the answer sheet/booklet provided.
2. Do **Not Write** on the Question Paper.

Question One (20 marks)

1. You are to design a home automation module that controls lights and fans using a microcontroller and actuators.
 - a) Distinguish between a sensor and an actuator and give 2 examples of each used in your system. (4 marks)
 - b) Briefly describe how a relay module can be used with a microcontroller to switch high-voltage AC appliances like fans. (6 marks)
2. Imagine you're developing an IoT system for monitoring environmental conditions in a smart farm.
 - a) List three types of sensors suitable for this task and briefly explain how each works. (6 marks)
 - b) Describe the role of a microcontroller in this system and how it interfaces with the sensors. (4 marks)

Question Two (10 marks)

1. Using Ohms law, solve the following problem
 - a) A light bulb has a resistance of $240\ \Omega$ and is connected to a 120V power supply, calculate the current flowing through the bulb. (3 marks)

- b) A toaster draws a current of 5A and has a resistance of $10\ \Omega$. Calculate the voltage across the toaster. (3 marks)
- c) A hairdryer operates at 220V and draws 11A of current.
- Calculate the resistance of the drier (2 marks)
 - If the resistance is increased to 30Ω , what would be the new current drawn by the hairdryer? (2 marks)